Model Complexity versus Transportability in Ovarian Cancer Risk Prediction

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***Abstract*—Preoperative blood tests help decide whether an ovarian mass requires cancer surgery; a model that degrades silently at a new hospital can delay surgery or trigger unnecessary operations. This paper asks whether more complex machine learning models transfer better to new hospitals than simple ones. Eight models (a CA125 cutoff rule, the clinical formulas ROMA and CPH-I, a logistic regression, an integer scorecard, and three tree ensembles) were trained on a Chinese dataset (349 patients) and tested, unchanged, on two other Asian cohorts (West China, 380 patients; Japan, 177). On the 100 West China patients with measured biomarkers, the scorecard (AUROC 0.971) matched the logistic regression (0.941) and ROMA (0.965); all three confidence intervals overlapped, and the simple model never performed worse. A computer simulation, calibrated to the real data, found that flexible models won only when the biomarker–cancer relationship stayed constant across hospitals; under concept drift, the simple scorecard won. We demonstrate that for small clinical datasets subject to real-world concept drift, a hand-computable integer scorecard provides a more transparent, readily auditable alternative to complex tree ensembles without sacrificing diagnostic accuracy.**

***Index Terms*—ovarian cancer; risk prediction; scorecard; external validation; transportability**

I. Introduction

Ovarian cancer is the most lethal gynecologic cancer: it causes over 200,000 deaths per year worldwide, most cases are found late, and screening has not reduced deaths. When a pelvic mass is found, the decision of whether to send the patient to a specialist cancer surgeon depends on preoperative risk, and getting it wrong in either direction is costly. That decision currently relies on two blood proteins that ovarian tumors release into the bloodstream; both are measured cheaply and routinely before surgery, and the open question is how to combine the two measurements into one trustworthy risk number. The markers are CA125 (cancer antigen 125) and HE4 (human epididymis protein 4), used either in fixed formulas such as the Risk of Ovarian Malignancy Algorithm (ROMA) [1] and the Copenhagen Index (CPH-I) [2], or in a fitted model. Machine learning models have been proposed for the same task, usually with strong results on the data they were trained on but little testing at other hospitals [3], [4].

This paper addresses two linked questions. First, does model complexity translate into performance that survives moving to a different hospital? Second, can a model simple enough to be computed at the bedside (an integer scorecard) match the more complex alternatives, and would a doctor be worse off using it? We trained eight models on one Chinese cohort and applied them without any change to two independent Asian cohorts, measuring discrimination, calibration, and decision consequences. We also built a computer simulation to explore *when* complex models help: they should win when the underlying biomarker–cancer relationship is stable, but may lose when that relationship drifts between settings, for example when assays or patient populations differ.

II. Methods

A. Data

Three public, anonymized datasets were used [17]–[19]: a training cohort from Changzhou, Jiangsu, China (349 patients; 171 ovarian cancers, 178 benign masses; Mendeley Data, doi:10.17632/th7fztbrv9.11, released with the study "Using machine learning to predict ovarian cancer"), a second Chinese cohort from the West China Second University Hospital of Sichuan University, Chengdu (380 patients; 188 cancers, 192 benign cysts; figshare, doi:10.6084/m9.figshare.28831256, collected 2024), and a Japanese cohort from a university hospital in Japan (177 patients; 27 cancers, 150 healthy women; Karger figshare, doi:10.6084/m9.figshare.24235450, 2016–2018). The two Chinese cohorts are surgical series of women with pelvic masses; the Japanese cohort contains healthy volunteers. Each patient has four features: CA125 (U/mL), HE4 (pmol/L), age, and menopausal status. In West China, HE4 was missing for 74% of patients; in Japan, age for 85%. Missing external values were replaced with the training-set median, computed once and frozen, so that no statistic is ever computed on a validation cohort. The Japanese cohort is a healthy-volunteer population (cancer prevalence 15%, 27 of 177), unlike the surgical cohorts (near 50%). It is therefore treated strictly as a specificity stress test, not as external validation of the triage tool: AUROC is reported separately, no decision-consequence analysis is performed there, and its metrics are never pooled with West China's to support clinical claims. The primary West China analyses use the 100 patients with measured HE4 (63 cancers, 37 benign), so that the external result rests on measured biology rather than imputed constants; the 280 excluded patients (125 cancers, 155 benign) had similar age, CA125, and menopausal rates. The 74%-imputed full cohort (N = 380) is reported alongside as a co-primary analysis.

B. Models

Eight models used the same four features. (1) *Integer scorecard*: each continuous feature was rank-transformed and cut into training-set tertiles (frozen cutpoints: CA125 24.3/165.0 U/mL, HE4 45.8/92.7 pmol/L, age 37.0/52.0 years), then one-hot encoded and fitted with an L2-regularized logistic regression ($C = 1.5$); each coefficient β_i was then rescaled to the nearest whole number $w_i = \text{round}(5 \times \beta_i / \max_j |\beta_j|) \in \{-5, \ldots, +5\}$, with $|\beta_i| < 0.01$ set to zero. The score is the sum of these feature points (range -7 to $+9$), can be computed by hand (Table I), and contains seven fitted integer parameters (two free bin weights for each of the three continuous features, plus one intercept that is not used in scoring). Its weights match pathophysiology: HE4 spans -4 to $+5$ versus -1 to $+2$ for CA125, matching HE4's higher cancer specificity and CA125's elevation in endometriosis; older age adds points. The configuration was selected among 12 candidates by repeated cross-validation (Section III-E). The regularized fit assigned the binary menopausal feature a coefficient that rounded to zero, so it is effectively dropped; it was not manually removed, and its row in Table I is retained to document this. (2) *Logistic regression* on standardized features. (3–4) the published formulas *ROMA* and *CPH-I*, computed on raw values as follows. ROMA uses $\text{PI} = -12.0 + 2.38 \ln(\text{HE4}) + 0.0626 \ln(\text{CA125})$ premenopause and $\text{PI} = -8.09 + 1.04 \ln(\text{HE4}) + 0.732 \ln(\text{CA125})$ postmenopause, mapped to a percentage by the logistic transform, with published cutoffs 13.1% (pre) and 27.7% (post) [1]. CPH-I uses $-14.0647 + 1.0649 \log_2(\text{HE4}) + 0.6050 \log_2(\text{CA125}) + 0.2672 \times \text{age}/10$, likewise logistic-transformed, with cutoff 7% [2]. Units: HE4 in pmol/L, CA125 in U/mL, age in years; menopause is the dataset's binary postmenopausal indicator; imputed HE4 uses the frozen training median. Within-cohort recalibration means fitting a logistic regression of the outcome on the formula's probability as the sole covariate, on West China. (5) a single *CA125 ≥ 35 U/mL* rule. (6–8) *XGBoost*, *CatBoost*, and *Random Forest*, evaluated under two settings: Setting A keeps each package's default hyperparameters (a deployment-ready, low-tuning comparison), and Setting B tunes each ensemble by nested cross-validation, with all tuning decisions confined to training folds (XGBoost 100 trees, depth 3; CatBoost 500 iterations, depth 4; Random Forest 500 trees, depth 6). The primary scientific comparison uses Setting B (Table II, Fig. 1); Setting A values are in the table footnote. All models were trained once on the Chinese cohort, and all parameters, bin cutpoints, and thresholds were frozen before any external evaluation; the models were applied unchanged everywhere else.

C. Evaluation

Discrimination was measured by the area under the ROC curve (AUROC), with 95% confidence intervals from 2,000 patient-level bootstrap resamples, two-sided paired bootstrap tests between models, and a DerSimonian–Laird random-effects model [9] pooling the two external cohorts. The pre-specified primary comparison is the scorecard versus ROMA (pooled); all other comparisons are secondary, and effect sizes with 95% CIs are emphasized over p-values. After Holm–Bonferroni correction across the five pooled pairwise comparisons in Section III-B, the primary comparison remained significant (adjusted $p = 0.012$). Calibration was measured with the Brier score and calibration-in-the-large. Decision value was quantified with a weighted-harm analysis (Section III-B): decision thresholds were swept over 1–100%, and for exchange rates $w \in \{1, 5, 10, 50\}$ (the harm of one missed cancer relative to one unnecessary referral) the threshold minimizing $H = w \times \text{missed} + \text{unnecessary}$ per 100 women was identified. The paper answers two distinct questions: Q1, whether each model transports unchanged (all discrimination analyses and the published-cutpoint comparison in Section III-B), and Q2, how well each model performs after local recalibration (calibration, DCA, and primary harm analyses). The exchange-rate weights are hypothetical scenarios, not validated clinical utilities: they span from valuing a missed cancer and an unnecessary referral equally ($w = 1$) to valuing a missed cancer 50 times as much ($w = 50$). These analyses are intended to characterize sensitivity to different relative error costs rather than to establish clinical utility. The primary comparison applies the same within-cohort recalibration to every model; published-cutpoint operating points are reported separately as a secondary, off-the-shelf comparison. Pooling was limited to discrimination metrics across the two external cohorts; no pooled decision analysis was performed. The retraining experiment used stratified training subsets of size $n \in \{60, 90, 120, 180, 240, 349\}$ (10 draws each), with models evaluated unchanged on the imputed West China cohort ($N = 380$). Robustness was tested with 10% and 30% multiplicative, additive, and combined measurement noise on CA125 and HE4, 5% gross-error contamination, and ±10% shifts of the scorecard's bin boundaries.

TABLE I
THE INTEGER SCORECARD, FITTED ON THE CHINESE TRAINING COHORT (349 PATIENTS) AND APPLIED UNCHANGED EXTERNALLY. FEATURES ARE BINNED BY FROZEN TRAINING-SET TERTILES; THE BINARY MENOPAUSAL FEATURE RECEIVED A COEFFICIENT THAT ROUNDED TO ZERO, SO IT CONTRIBUTES NO POINTS.

Feature	Low bin	Mid bin	High bin
CA125 (U/mL)	−1	0	+2
HE4 (pmol/L)	−4	−1	+5
Age (years)	−2	0	+2
Menopausal status	0	0	0

D. Implementation and availability

Analyses used Python 3.13 with scikit-learn 1.7.2, XGBoost 3.3.0 (100 trees, depth 6, learning rate 0.3), CatBoost 1.2.10 (1,000 iterations, depth 6, learning rate 0.03), and Random Forest (100 trees, no depth cap), all with random seed 42; cross-validation used stratified five-fold splits, and the simulation used 20 replicates per condition. The complete code, figure scripts, and results are available at https://github.com/angelayaoo/Ovarian-Cancer-RiskSLIM, with a one-command pipeline (`reproduce.sh`) and pinned dependencies.

E. Simulation

The simulation is an illustrative counterexample, not an explanation of the observed real-world results: complexity can hurt under certain forms of concept drift; it varies drift (d) independently of covariate shift (s) and noise, leaving other drift forms (prevalence, scale, nonlinear-to-linear) to future work. A simulation calibrated to the Chinese data generated synthetic patients: biomarkers were drawn from class-conditional lognormal distributions, and the cancer label followed the logistic model

$$\begin{aligned}\eta(z_1, z_2, z_3) = & -0.65 + 1.8z_1 + 1.0z_2 + 0.5z_3 \\ & + (1 - d)\big(1.6z_1z_2 + 1.2\sin(1.7z_1)\big),\end{aligned} \tag{1}$$

where z_1, z_2, and z_3 are the standardized log CA125, log HE4, and age, respectively, and $d \in [0, 1]$ sets the drift level: $d = 0$ keeps the interaction and nonlinear terms, $d = 1$ removes them from the test-side model, and intermediate values scale them by $1 - d$, as if the biomarker–cancer relationship partially changed between hospitals. The grid crossed training sizes $n \in \{100, 200, 400, 800\}$, distribution shifts $s \in \{0, 0.5, 1.0\}$ (shrinking class-conditional biomarker means toward the pooled mean), measurement noise $\{0\%, 50\%\}$, and drift levels $d \in \{0, 1\}$, with 20 replicates per cell.

III. Results

A. Discrimination

On the 100 West China patients with measured HE4 (63 cancers), the scorecard scored 0.971, ROMA 0.965, tuned XGBoost 0.946, and the logistic regression 0.941, with overlapping intervals; the remaining models were lower (Table II, West CC column; Fig. 1).

On the co-primary 74%-imputed full cohort ($N = 380$), the ranking was similar (logistic regression 0.936, scorecard 0.929, tuned ensembles 0.877–0.895; Table II, West column).

As a discrimination benchmark, a random-effects (DerSimonian–Laird) meta-analysis with paired bootstrap tests across the two external cohorts gave a pooled scorecard-minus-ROMA difference of +0.043 AUROC (95% CI 0.013 to 0.074; $p = 0.006$). The scorecard also exceeded every tree ensemble (all p at most 0.002), and it did not differ from the logistic regression (−0.005; $p = 0.55$). After Holm–Bonferroni correction across these five comparisons, the scorecard–ROMA difference remained significant (adjusted $p = 0.012$), as did all ensemble comparisons (adjusted p at most 0.010).

In the Japan specificity stress test ($n = 177$), CPH-I was highest (0.920; Table II, Japan column), and the pooled CPH-I comparison was heterogeneous ($I^2 = 79\%$). Within Japan, 86% of healthy women but only 30% of cancer patients fell into the lowest bins of both markers.

Per-feature Kolmogorov–Smirnov shifts from the training distribution averaged 0.22 (West China) and 0.50 (Japan). The West China cohort had higher CA125 (median 65.1 versus 49.3 U/mL) than the training patients, and the Japanese volunteers had far lower marker levels (CA125 15.1 U/mL, HE4 33.8 pmol/L).

TABLE II

AUROC (95% CI) by model and cohort, bootstrap means over 2,000 resamples. West CC: the 100 West China patients with measured HE4 (primary external benchmark); West: the full 74%-imputed cohort (N = 380, co-primary analysis); Japan: healthy volunteers (specificity stress test). Setting A (defaults) ensembles scored 0.921/0.922/0.910 (CC), 0.887/0.879/0.879 (West), 0.734/0.749/0.792 (Japan). Bold: highest value per cohort, or all values statistically indistinguishable from it (paired bootstrap).

Model	West CC (n=100)	West (N=380)	Japan
Scorecard	**0.971** **(0.936–0.995)**	**0.929** **(0.902–0.955)**	0.845 (0.714–0.949)
Logistic reg.	0.941 (0.887–0.983)	**0.936** **(0.908–0.961)**	0.806 (0.666–0.926)
ROMA	0.965 (0.930–0.992)	0.886 (0.848–0.921)	0.799 (0.666–0.908)
CPH-I	0.929 (0.875–0.973)	0.868 (0.828–0.905)	**0.920** **(0.852–0.973)**
CA125 rule	0.694 (0.605–0.785)	0.642 (0.597–0.684)	0.781 (0.685–0.874)
XGBoost	0.946 (0.897–0.984)	0.890 (0.853–0.925)	0.840 (0.711–0.944)
CatBoost	0.910 (0.845–0.962)	0.877 (0.836–0.915)	0.772 (0.650–0.880)
Random Forest	0.930 (0.871–0.979)	0.895 (0.858–0.931)	0.819 (0.692–0.926)

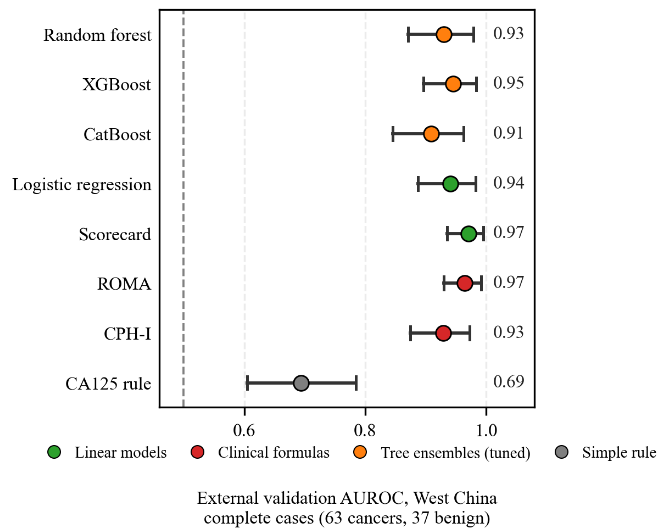

Fig. 1. External validation AUROC, 100 West China patients with measured HE4 (63 cancers). Dots: bootstrap means with 95% CIs (2,000 resamples). Models ordered by increasing complexity. Dashed line: AUROC 0.5 (chance).

B. Decision consequences

In the primary weighted-harm analysis (Q2: post-recalibration performance; Table III, Figure 2), every model was within-cohort recalibrated. The scorecard's minimum harm per 100 women was below recalibrated ROMA and CPH-I at weights up to 5 (at $w = 1$: 12.1, versus 14.7 and 15.0). At $w = 10$ it remained slightly below recalibrated ROMA (45.0 versus 45.8). At $w = 50$, both recalibrated formulas fell below the scorecard (ROMA 45.8, CPH-I 50.3, versus 59.5). The logistic regression was lowest at every weight (9.7 to 46.6).

As the Q1 decision consequence (unchanged transport), at its high-risk tier (score $\geq$ +1) the scorecard missed 47 of 188 cancers with 5 false-positive referrals, while the published

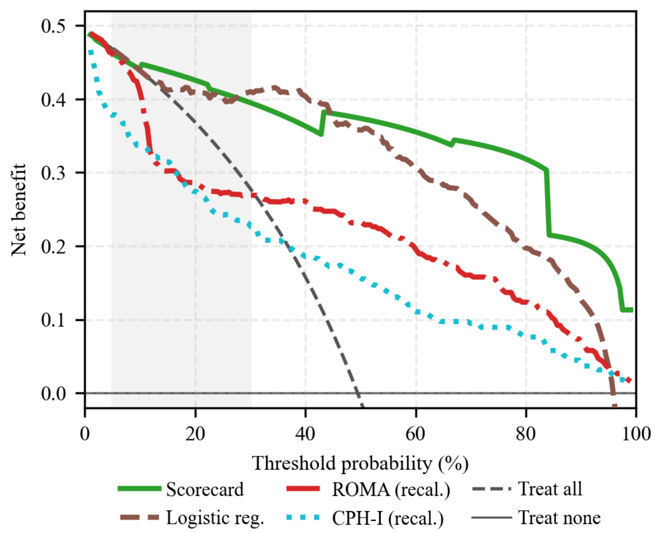

Fig. 2. Decision curve analysis on West China (N = 380): net benefit per patient (true positives minus threshold-weighted false positives) against the decision threshold in percent, with "treat all" (dashed) and "treat none" (the horizontal axis) as references; the shaded band marks the published-cutpoint triage range (7–30%). The scorecard and the two formulas are shown after within-cohort recalibration. XGBoost is omitted; uncertainty bands are omitted; minimum harms are in Table III.

TABLE III

Minimum net harm per 100 women on West China (N = 380) at each exchange-rate weight w (harm of one missed cancer relative to one unnecessary referral; harm = $w\times$missed+unnecessary). Lower is better. The scorecard, ROMA, and CPH-I are within-cohort recalibrated; the logistic regression and XGBoost are shown as fitted. Published-scale harms: ROMA 15.0/43.4/45.8/45.8; CPH-I 15.0/51.3/64.7/159.5.

Model	w=1	w=5	w=10	w=50
Logistic reg.	**9.7**	**29.2**	**39.7**	**46.6**
Scorecard	12.1	30.5	45.0	59.5
ROMA (recalibrated)	14.7	42.9	45.8	**45.8**
CPH-I (recalibrated)	15.0	50.3	50.3	50.3
XGBoost	14.7	44.7	66.1	215.5

cutpoints missed 82 (ROMA) and 48 (CPH-I) of 188 cancers.

C. Calibration and missing-data checks

For calibration on the imputed West China cohort (N = 380), the recalibrated scorecard had Brier 0.097, a probability-scale calibration slope of 1.00, and expected calibration error (ECE) of 0.042; the logistic regression had Brier 0.111, slope 1.44, and ECE 0.135. The published formulas were farther from the data: ROMA had Brier 0.209, slope 1.53, and ECE 0.220; CPH-I had Brier 0.259, slope 1.11, and ECE 0.303. Both over-predicted risk (calibration-in-the-large −0.95 and −1.42).

Refitting without HE4 gave 0.925 (scorecard), 0.932 (logistic regression), and 0.891 (XGBoost) on the imputed cohort, agreeing with the complete-case ranking (Table II). Multiple imputation by chained equations (MICE) preserved the flexible-model ranking (logistic regression 0.926) but degraded the scorecard's tertile binning (0.853), motivating the complete-case primary analysis.

D. Robustness and simulation

For robustness to measurement noise (West China, N = 380), losses under 30% multiplicative noise were 0.040 for the scorecard, 0.002 for the logistic regression, 0.074 for ROMA, and 0.086 for XGBoost.

In the simulation (Figure 3), the ensemble-average minus-scorecard gap averaged +0.018 at drift level d = 0 of (1) and −0.018 at d = 1; at d = 1 the scorecard had the highest AUROC in 22 of 24 cells. Across the four simulation factors, the spread of the gap was largest across drift levels (0.033), followed by distribution shift (0.012), training size (0.003), and noise (0.002).

E. Internal validity and sensitivity

Across 20 repeated five-fold cross-validations, the scorecard's internal AUROC was 0.906 (SD 0.003), and the deployed three-bin configuration was selected in 18 of 20 repeats. In-sample the ensembles fitted at 0.991–1.000, falling 0.094–0.117 under CV.

IV. Discussion and Conclusion

The results show three things. First, on discrimination: adding model complexity did not buy external accuracy. Seven integer parameters matched models with thousands of parameters; the scorecard was numerically best on the complete cases (0.971), the logistic regression matched or slightly exceeded it on the imputed cohort (0.936 vs. 0.929), and CPH-I won the Japan stress test (0.920). Second, on decisions: the published cutpoints missed 82 of 188 cancers versus 47 for the locally fitted scorecard, and with all models recalibrated its minimum harm stayed lowest up to 5:1. Third, on why: the simulation shows one concrete theoretical mechanism by which complexity can hurt when the training structure does not persist.

In short, a hand-computable integer scorecard can match far more complex models, miss fewer cancers than off-the-shelf formulas, and stay fully transparent.

The main limitations are as follows. (i) All three cohorts are Asian hospital series, so results may not transfer to other populations. (ii) the 74% missing-HE4 rate in West China and the small Japanese sample (n = 177) are structural limitations, but complete-case and imputed results agreed. (iii) ultrasound-based models could not be tested (the public data lack imaging); (iv) the simulation is a simplification, not a proof about any specific hospital.

Future work should add ultrasound variables, validate prospectively, and test other tumor markers. As a proof of concept for transparent triage (retrospective, not prospective validation), the integer scorecard offers a potentially auditable, computationally free alternative that matches the best-performing models while remaining computable by hand.

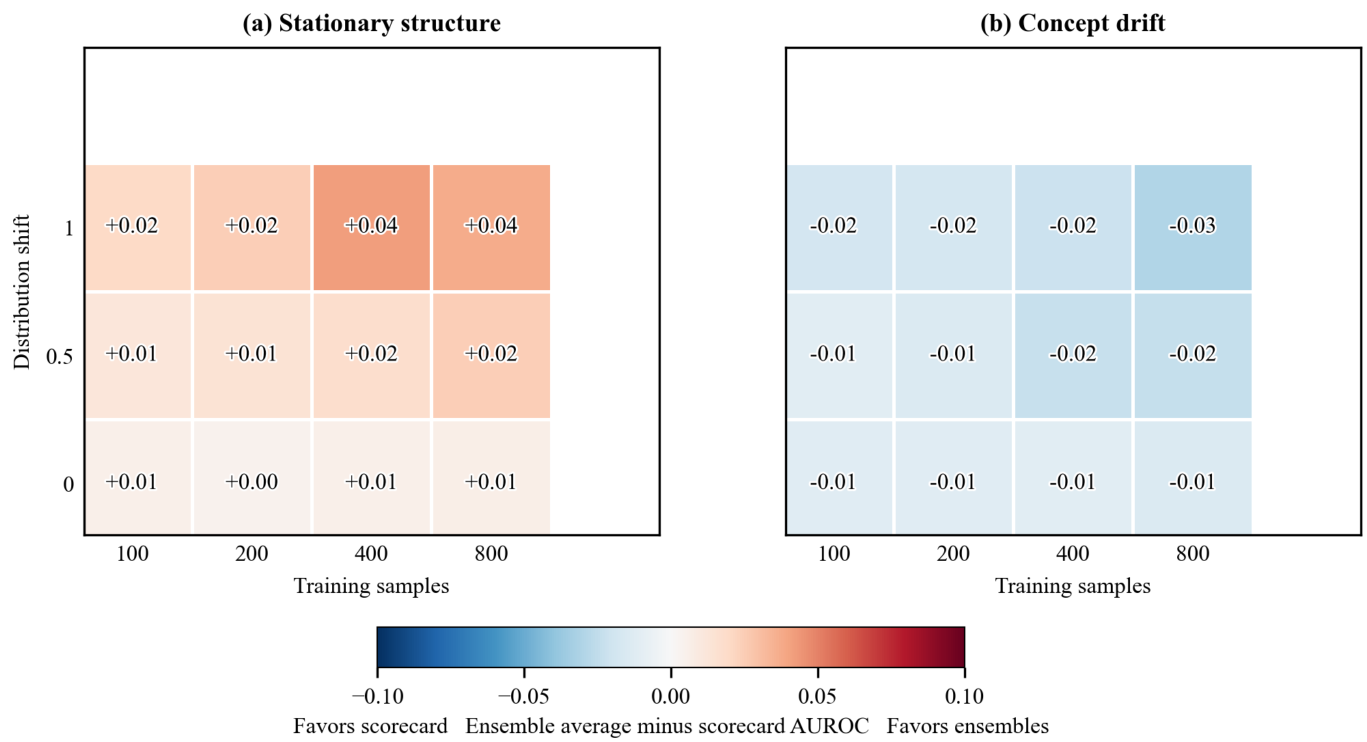

Fig. 3. Regime map from the simulation: each cell is the mean over 20 replicates of the tree-ensemble-average minus scorecard test AUROC at 50% noise; red favors the ensembles, blue the scorecard. Axes: training samples $n \in \{100, 200, 400, 800\}$; distribution shift $s \in \{0, 0.5, 1.0\}$.

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Appendix A. Glossary of Terms

AUROC: area under the receiver operating characteristic curve; CA125: cancer antigen 125 (U/mL); HE4: human epididymis protein 4 (pmol/L); ROMA: Risk of Ovarian Malignancy Algorithm; CPH-I: Copenhagen Index; CI: confidence interval; CV: cross-validation; ECE: expected calibration error; DCA: decision curve analysis.